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(54) **SMALL-SIZED FAST COLD AND HOT SHOCK TEST DEVICES**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,184,504 B1 2/2001 Cardella
2003/0155504 A1 8/2003 Motchkine et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CA 1038912 A 9/1978
CN 102614944 A 8/2012
(Continued)

OTHER PUBLICATIONS

International Search Report in PCT/CN2021/130590 mailed on Mar. 16, 2022, 9 pages.

(Continued)

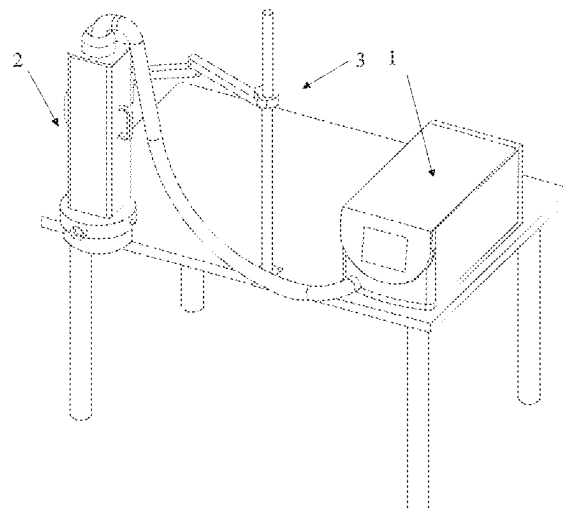
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(57) **ABSTRACT**

A small-sized fast cold and hot shock test device is provided. The device includes a host, a test head used for cold and hot shock to a component under test by temperature control and output of compressed air, and an adjustment device for adjusting a position of the test head. The host includes a control device at least used to send temperature control data to the test head. The test head includes an eddy current mechanism for cooling or heating the compressed air, a heater including an air inlet end and an air outlet end, and a nozzle. The eddy current mechanism includes an air inlet, a cold air outlet, and a hot air outlet. The air inlet is connected with an air supply system through an intake air line unit. The cool air outlet is connected with the air inlet end of the heater.

12 Claims, 6 Drawing Sheets



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(56)

References Cited

U.S. PATENT DOCUMENTS

2004/0051545 A1* 3/2004 Tilton G05D 23/1919
324/750.08
2006/0244472 A1* 11/2006 Hayashi G01R 31/2874
324/750.08
2009/0158975 A1 6/2009 Cluzel et al.
2010/0039117 A1 2/2010 Jacobs
2016/0091559 A1 3/2016 Teoh
2021/0011075 A1 1/2021 Zou

FOREIGN PATENT DOCUMENTS

CN 203413834 U 1/2014
CN 104564478 A 4/2015
CN 204788951 U 11/2015
CN 205332585 U 6/2016
CN 105841936 A 8/2016
CN 106918524 A 7/2017
CN 206378266 U 8/2017
CN 107300264 A 10/2017
CN 107703240 A 2/2018
CN 206998703 U 2/2018

CN 108645739 A 10/2018
CN 111487156 A 8/2020
CN 111913099 A 11/2020
CN 111948082 A 11/2020
CN 113203913 A 8/2021
DE 3720259 A1 12/1988
JP 2000356498 A 12/2000
KR 20200112345 A 10/2020

OTHER PUBLICATIONS

Written Opinion in PCT/CN2021/130590 mailed on Mar. 16, 2022, 11 pages.
First Office Action in Chinese Application No. 202110760705.3 mailed on Aug. 17, 2021, 17 pages.
Zhang, Haifeng, The Design of Diesel Engine Thermal Shock Test Bed, Internal Combustion Engines, 37-40, 2013.
Dong, Miao et al., Research of Thermal Shock Reliability Tests for Gasoline Engine, Small Internal Combustion Engine and Vehicle Technique, 45(6): 27-29, 2016.
Vladimir N. Zudnov et al., The Interaction of Streamwise Vortex With a Shock Wave, 20th International Congress on Instrumentation in Aerospace Simulation Facilities, 2003. ICIAASF '03., 258-267, 2003.
Henri Malinen et al., Optimizing the Shock Wave-to-Vortex Power Ratio at the End of a Waveguide, 2019 IEEE International Ultrasonics Symposium (IUS), 2019, 4 pages.

* cited by examiner

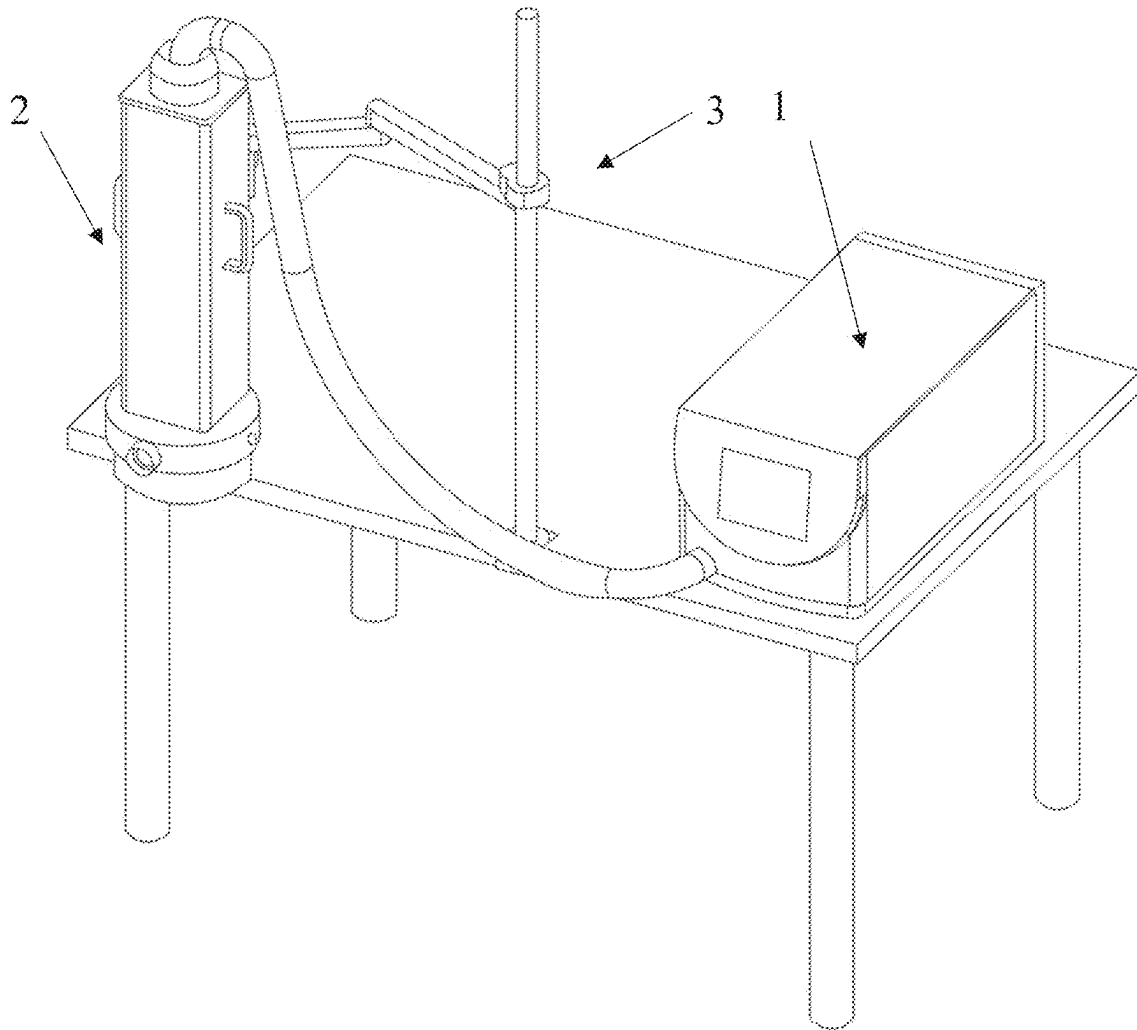


FIG. 1

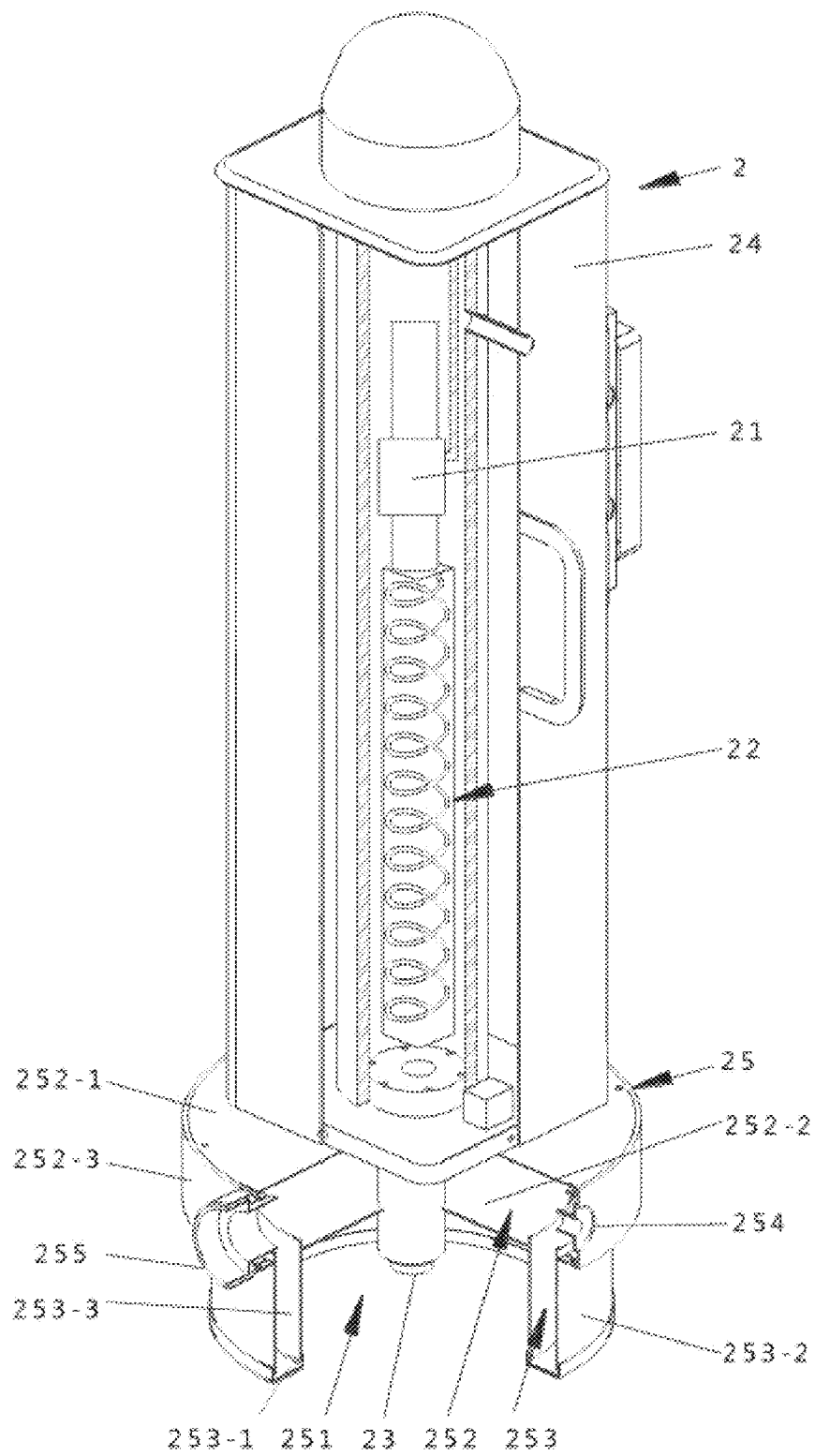


FIG. 2

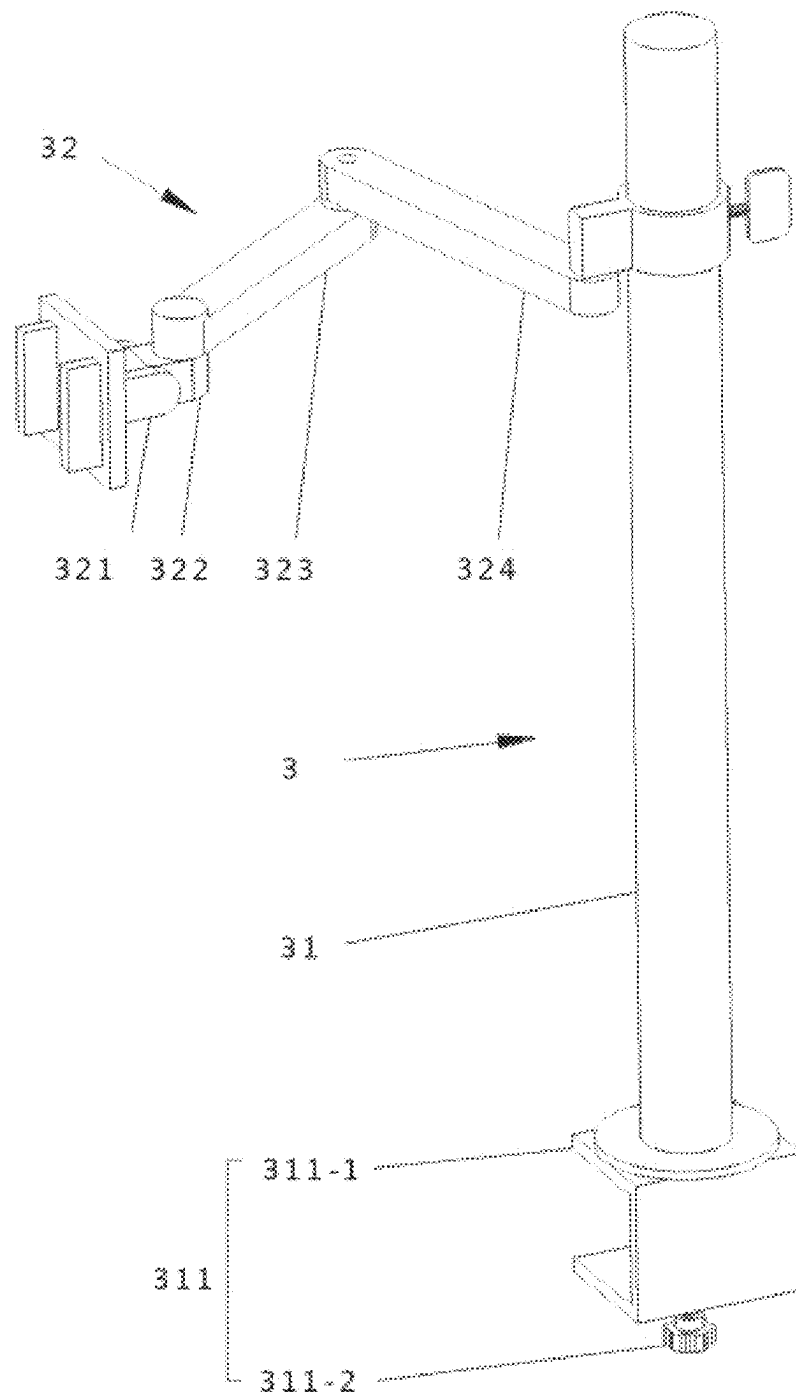


FIG. 3

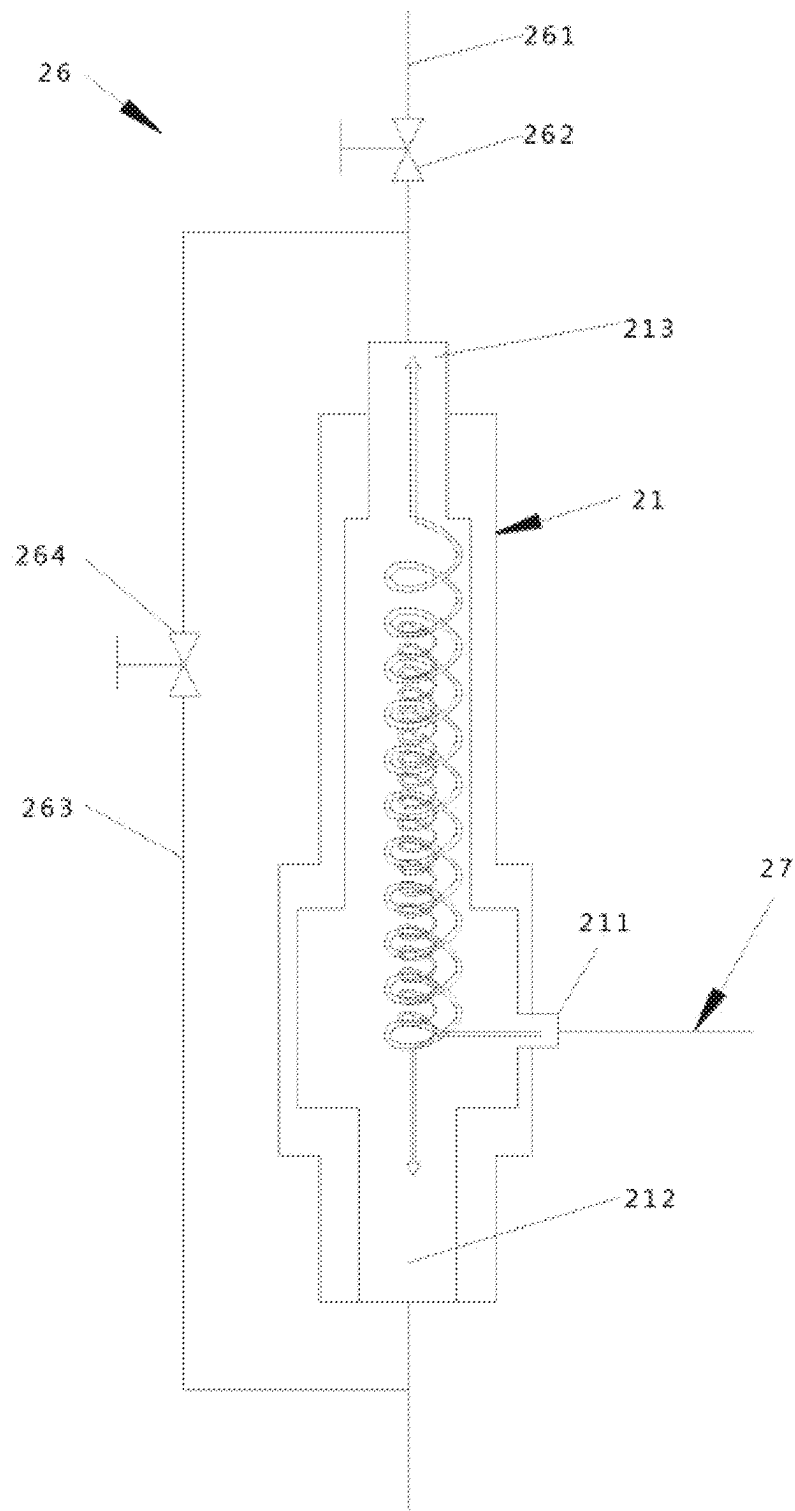


FIG. 4

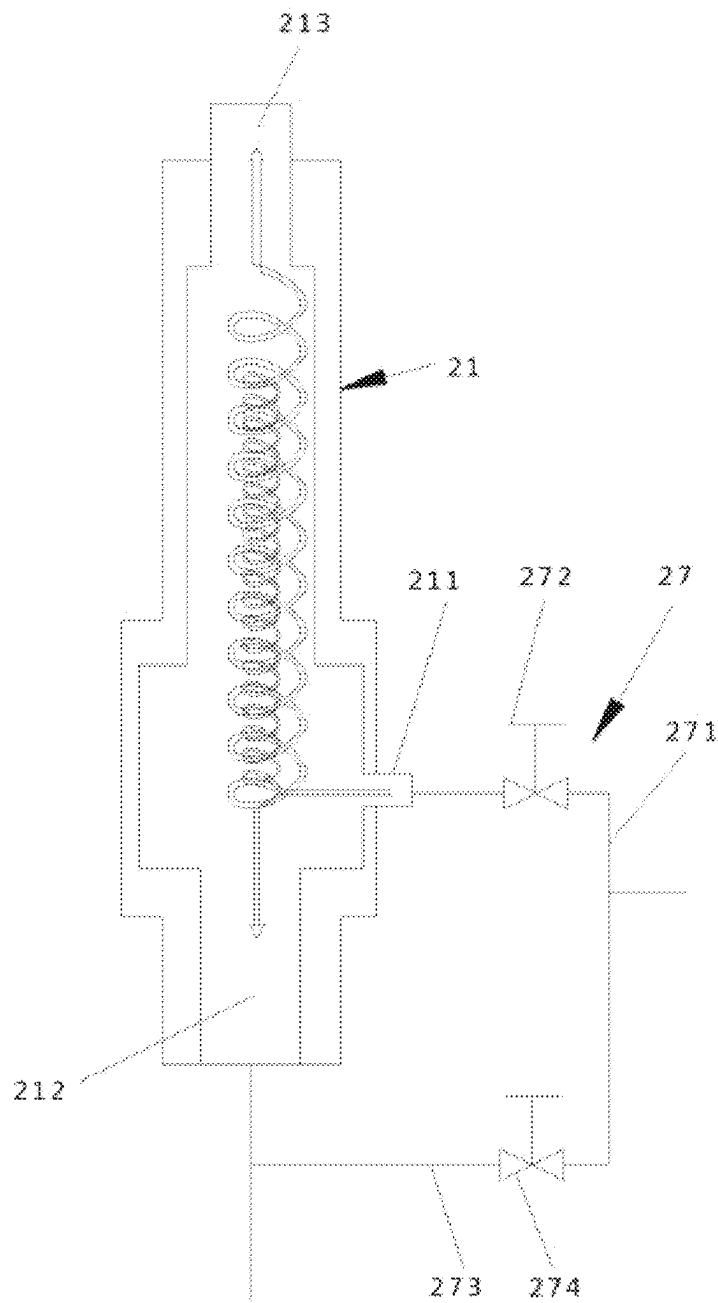


FIG. 5

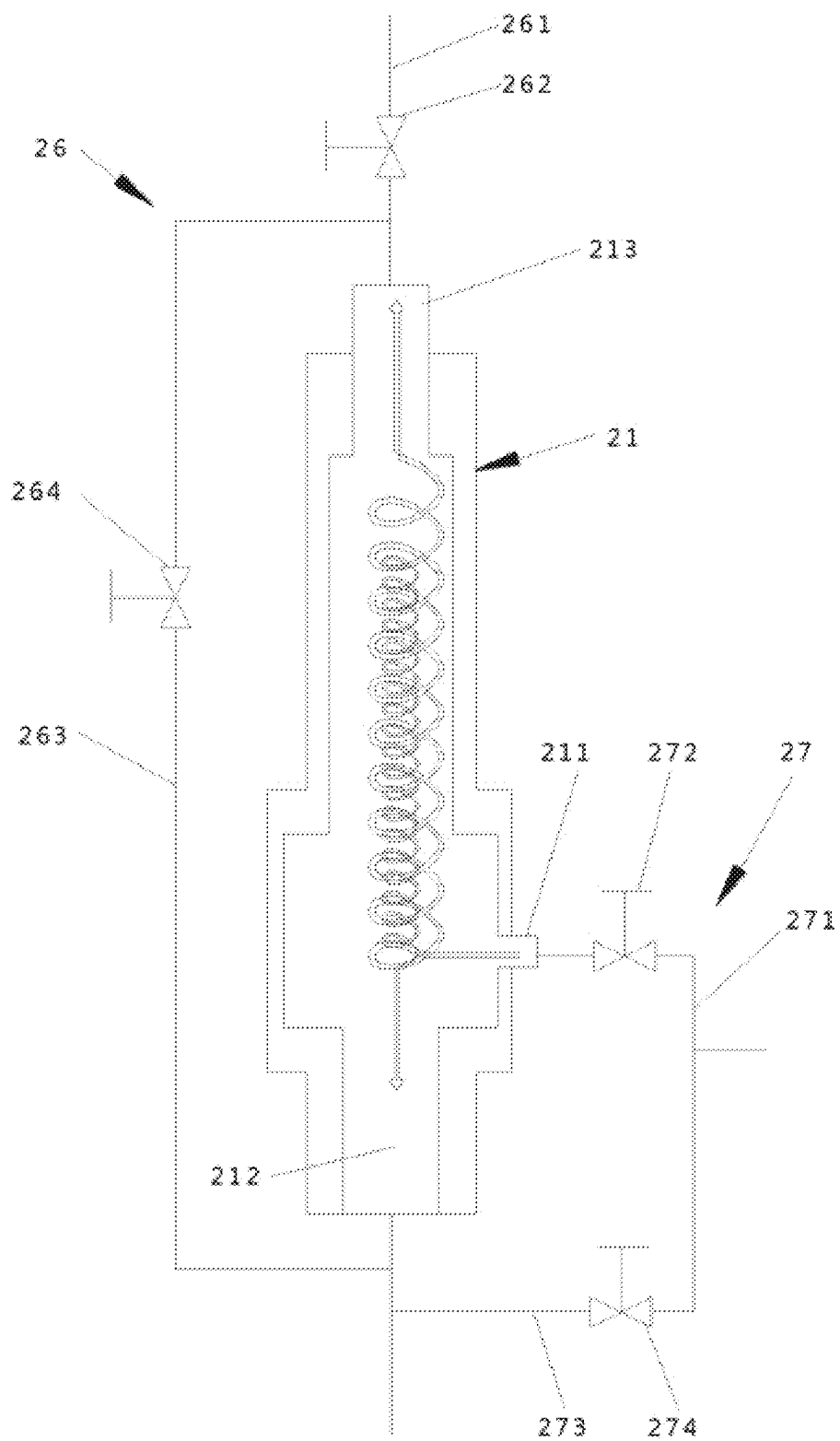


FIG. 6

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SMALL-SIZED FAST COLD AND HOT SHOCK TEST DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 18/534,693, filed on Dec. 10, 2023, which is a continuation of International Application No. PCT/CN2021/130590, filed on Nov. 15, 2021, which claims priority to Chinese Patent Application No. 202110760705.3, filed on Jul. 6, 2021, the contents of each of which are entirely incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to the field of components testing equipment, and in particular, to a small-sized fast cold and hot shock test device.

BACKGROUND

In the manufacturing process of semiconductors and components, especially chip products, performance testing or reliability testing is needed, including chemical changes or physical damage caused by extremely high temperature and extremely low temperature continuous environment, thus, a cold and hot shock testing machine is required.

Fast heating and cooling rate, wide range of test temperature, low energy consumption have always been the demands of improvement of hot and cold shock testing machine. However, the common cold and hot shock test devices are refrigerated by compressor, evaporator, and condenser, and then the output temperature (roughly between -40°C . and 200°C .) is adjusted through the heating wire. This kind of structure and cooling and heating methods of cold and hot shock test device have the advantages of slow heating speed, high energy consumption, and high cost, as a result, the device is bulky and cannot be moved according to the site, and a component under test can only be placed in a fixed position on the test device, but the test device cannot be positioned with the component, which lacks flexibility and convenience.

SUMMARY

One or more embodiments of the present disclosure provide a small-sized fast cold and hot shock test device. The device includes a host, a test head used for cold and hot shock to a component under test by temperature control and output of compressed air, and an adjustment device for adjusting a position of the test head. The host includes a control device at least used to send temperature control data to the test head. The test head includes an eddy current mechanism for cooling or heating the compressed air, a heater including an air inlet end and an air outlet end, and a nozzle. The eddy current mechanism includes an air inlet, a cold air outlet, and a hot air outlet. The air inlet is connected with an air supply system through an intake air line unit. The cold air outlet is connected with the air inlet end of the heater. The nozzle is connected with the air outlet end of the heater.

In some embodiments, the test device is supplied with the compressed air by the air supply system. The compressed air is passed through the host first, and then input into the eddy current mechanism of the test head, which makes the airflow supplied to the eddy current mechanism more stable. The air

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supply system mentioned here can be the equipment or system that can provide compressed air such as the central air compression system at the production and manufacturing site. The compressed air simultaneously generates a cold air flow and a hot air flow after passing through the eddy current mechanism, the temperature of the cool air and the hot air is fixed, and the cool air is fed into the heater from the cold air outlet, and is compensated by the heater to adjust the temperature to the required temperature for testing.

In some embodiments, the test device replaces a compressor, evaporator, condenser, and other traditional refrigeration equipment by setting the eddy current device. The test device replaces the compressor, evaporator, condenser, and other traditional refrigeration equipment by setting eddy current device, which makes a test temperature range wide and a speed of heating and cooling fast. The test temperature range is -55°C . to 220°C ., which takes about 1 min for the temperature to reduce from 220°C . to -55°C . and takes about 20 s for the temperature to rise from -55°C . to 220°C ., and for the most commonly used temperature in the test, for example, it only takes about 12 s for the temperature to reduce from 70°C . to 0°C ., and only takes about 20 s for the temperature to reduce from 85°C . to -40°C ., which greatly improves the efficiency compared with the existing cold and hot shock test device. At the same time, the eddy current mechanism does not need electricity for heating and cooling compressed air, which greatly reduces energy consumption compared with the existing cold and heat shock test device equipped with the compressor.

In some embodiments, the test device eliminates the compressor and semiconductor refrigeration sheet, thus reducing an overall volume of the device to make it easy to move. The test head is designed to be suitable for situations where selective testing is required, for example, a situation where a component on a circuit board is tested without causing damage to other surrounding components.

In some embodiments, the adjusting device is used to adjust the position of the test head, which changes the structure that the test head is fixedly connected to the test device. In practice, the adjusting device is not attached to the test device and can be installed in any position such as table top, floor, etc. By changing the position of the adjusting device, a test area covered by the test head movement is enlarged to a certain extent, and the operation convenience and mobility flexibility of the test device are effectively improved.

In some embodiments, a bypass line unit is arranged between the eddy current mechanism and the heater. The bypass line unit includes a first line with one end connected with the hot air outlet, a first solenoid valve disposed on the first line, a second line with one end connected with the first line and the other end connected with the air inlet end of the heater, and a second solenoid valve disposed on the second line. The connection of the one end of the second line on the first line is between the first solenoid valve and the air inlet end of the heater.

In some embodiments, in a case of the bypass line unit being arranged, the first solenoid valve is opened and the second solenoid valve is closed when the small-sized fast cold and hot shock test device is in a cooling working state. At this time, hot air generated by the eddy current mechanism is output from the hot air outlet, and cold air is output from the cold air outlet and into the heater downstream. The first solenoid valve is closed and the second solenoid valve is opened when the small-sized fast cold and hot shock test device is in a heating working state. At this time, the cold air generated by the eddy current mechanism is output from the

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cold air outlet and into the heater downstream, and the hot air enters the heater downstream through the second line. A final function of the hot air entering the heater is to compensate the temperature of compressed air and reduce the heating burden of the heater, thus, further improving the heating efficiency and reducing the energy consumption of the heater.

In some embodiments, the intake air line unit is further connected with the air inlet end of the heater. Whether the compressed air is fed into the eddy current mechanism and the heater is switched through the intake air line unit. In particular, the intake air line unit includes a third line with one end connected with the air inlet and the other end connected with the air supply system, a third solenoid valve disposed on the third line, a fourth line with one end connected with the air inlet end of the heater and the other end connected with the air supply system, and a fourth solenoid valve disposed on the fourth line.

In some embodiments, when the small-sized fast cold and hot shock test device is in the cooling working state and the heating working state, the third solenoid valve is opened and the fourth solenoid valve is closed. That is, the eddy current mechanism always works, and the air supply system does not directly supply air to the heater. Further, the small-sized fast cold and hot shock test device has a high temperature heating state. When the small-sized fast cold and hot shock test device is in the high temperature heating state, the fourth solenoid valve is opened, and the first solenoid valve, the second solenoid valve, and the third solenoid valve are all closed. That is, the eddy current mechanism does not work, the air supply system directly supplies air to the heater, and the compressed air enters the heater directly for heating treatment through the fourth line.

In some embodiments, the heating working state and the high temperature heating state can be selected freely according to test requirements. When the required air for test is hot air at a lower target temperature, the first heating working state can be selected. The cold air and the hot air generated by the eddy current mechanism both enter the heater. Due to the lower target temperature, the burden of the heater by heating the cold air is light, at the same time, the hot air compensates the temperature, which further improves the heating speed and reduces the energy consumption of the heater.

In some embodiments, when the required air for test is hot air at a relatively high target temperature, the high temperature heating state can be selected. The compressed air enters the heater to be heated. At this time, no compressed air enters the eddy current mechanism, so that no cold air is generated. The compressed air is heated to the target temperature from normal temperature, which reduces the heating burden of the heater and saves energy.

In some embodiments, the intake air line unit is arranged among the eddy current mechanism, the heater, and the air supply system. The intake air line unit includes the third line with one end connected with the air inlet and the other end connected with the air supply system, the third solenoid valve disposed on the third line, the fourth line with one end connected with the air inlet end of the heater and the other end connected with the air supply system, and the fourth solenoid valve disposed on the fourth line.

In some embodiments, the small-sized fast cold and hot shock test device has the cooling working state and the heating working state. When the small-sized fast cold and hot shock test device is in the cooling working state, the third solenoid valve is opened and the fourth solenoid valve is closed, the compressed air enters the eddy current mechanism

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through the third line, the hot air generated by the eddy current mechanism is output from the hot air outlet, and the cold air is output from the cold air outlet and into the heater downstream. When the small-sized fast cold and hot shock test device is in the heating working state, the third solenoid valve is closed and the fourth solenoid valve is opened, and the compressed air enters the heater directly for heating treatment through the fourth line. At this time, the compressed air entering the heater is normal temperature, it takes less time and consumes less energy than the heater heat up the compressed air to the target temperature than that the heater to heat up the cold air to the target temperature.

In some embodiments, a temperature sensor is set in the nozzle for sensing the temperature of the compressed air entering the nozzle from the heater and transmitting sensing data to the control device.

In some embodiments the test head includes a shell that the eddy current mechanism and the heater are set in.

In some embodiments, the test head includes a cover set at a bottom of the shell. The cover is used to cover the component under test to form a closed space isolated from external air during the test to make the air used for shock more concentrated and the ambient temperature in the closed space closer to the required temperature for testing, which enables more accurate control of the temperature of the test environment and is conducive to energy saving.

In some embodiments, the cover has a cavity for accommodating the component under test during the test, a first cavity above the cavity, and a second cavity around an outside of the cavity. An outer part of the first cavity and an upper part of the second cavity are connected to form a sealing space sealed and isolated from the cavity. The sealing space has an inlet for air input and an outlet for air output, and the cavity has an air outlet for air output. The nozzle passes through the first cavity and a bottom of the nozzle extends into the cavity.

Frost or mist may form on the wall of the cavity when the extremely low temperature airflow meets the external normal temperature airflow during the cold and hot shock test, which is not conducive to an operator observing the real-time test situation of the component under test. Therefore, the inlet and the outlet are set in order to continuously input air into the sealing space during the test, which makes the air flow in the second cavity to form an air curtain and effectively prevents the condensation of water mist and frost on the wall of the cavity under the action of air flow, so that it is convenient for the operator to observe a state of the component under test directly, and the test can be carried out normally.

In some embodiments, the first cavity is formed by connecting a first body, a second body, and a third body. The first body is located directly above the second body, and an upper end and a lower end of the third body are connected with the first body and the second body, respectively. The second cavity is formed by connecting a gasket, a fourth body, and a fifth body. An upper end and a lower end of the fourth body are connected with a lower end of the second body and the gasket, respectively, and an upper end and a lower end of the fifth body are connected with the lower end of the second body and the gasket, respectively. The cavity is formed by connecting the second body and the fifth body. The inlet and the air outlet are both set on the third body. The gasket is set to make an interior of the cover, including the cavity and the sealing space formed by the first cavity and the second cavity, more enclosed. The shell is fixedly connected with the first body of the cover.

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In some embodiments, the adjustment device includes a fixing bracket and an adjusting bracket capable of being expanded and folded, and one end of the adjusting bracket is connected with the fixing bracket, and the other end of the adjusting bracket is slidably connected with the test head. Being expanded and folded of the adjusting bracket enables the test head to move over a wider area. Furthermore, the adjusting bracket includes a sliding block connected with the test head, a connecting block rotationally connected with the sliding block, a first connecting arm rotationally connected with the connecting block, and a second connecting arm rotationally connected with the first connecting arm and the fixing bracket. A bottom of the fixing bracket is provided with a clamping mechanism for fixing a position of the adjusting device. The clamping mechanism is separably connected to other objects. The clamping mechanism can be clamped on the table or other objects, and can also be clamped on the strip or plate shaped components of other devices, which is not only suitable for a plurality of objects, but also can be repeatedly disassembled.

In some embodiments, the sliding connection between the test head and the sliding block enables the test head and the sliding block to slide along a length direction of the test head. That is, a relative height of the test head and the adjusting bracket can be adjusted. An angle of the sliding block relative to the connecting block is adjusted through the rotary connection between the sliding block and the connecting block, so as to change the angle of the test head relative to the adjusting bracket. The rotary connection of the first connecting arm and the second connecting arm realizes the expansion and folding of the adjusting bracket, and increases the test area that covered by the test head movement. The second connecting arm is rotationally connected with the fixed bracket, so that the adjusting bracket can arbitrarily change its direction around the fixing bracket in the case that the fixing bracket has been connected to other objects.

The present disclosure at least has the beneficial effects as follows.

In some embodiments of the present disclosure, the test device replaces the compressor, evaporator, condenser, and other traditional refrigeration equipment by setting the eddy current device, which makes the test temperature range wide and the speed of heating and cooling fast. The test temperature range is -55°C . to 220°C ., which takes about 1 min for the temperature to reduce from 220°C . to -55°C . and takes about 20 s for the temperature to rise from -55°C . to 220°C ., and for the most commonly used temperature in the test, for example, it only takes about 12 s for the temperature to reduce from 70°C . to 0°C ., and only takes about 20 s for the temperature to reduce from 85°C . to -40°C ., which greatly improves the efficiency compared with the existing cold and hot shock test device. At the same time, the eddy current mechanism does not need electricity for heating and cooling compressed air, which greatly reduces energy consumption compared with the existing cold and heat shock test device equipped with the compressor. In addition, the test device eliminates the compressor and semiconductor refrigeration sheet, thus, reducing the overall volume of the device to make the device easy to move. The test head is designed to be suitable for situations where selective testing is required. The setting of the adjusting device effectively improves the operation convenience and mobility flexibility of the test device.

BRIEF DESCRIPTION OF THE DRAWINGS

Some specific embodiments of the present disclosure will be described in detail below by way of illustration and not

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limitation with reference to the accompanying drawings. Identical reference numerals in the drawings indicate identical or similar components or parts. It should be understood by those skilled in the art that these drawings are not necessarily drawn to scale, wherein:

FIG. 1 is a structural schematic diagram of a small-sized fast cold and hot shock test device according to some embodiments of the present disclosure;

FIG. 2 is a structural schematic diagram of a section structure of a test head according to some embodiments of the present disclosure;

FIG. 3 is a structural schematic diagram of an adjusting device according to some embodiments of the present disclosure;

FIG. 4 is a schematic diagram of an eddy current mechanism according to some embodiments of the present disclosure;

FIG. 5 is a schematic diagram of another eddy current mechanism according to some embodiments of the present disclosure; and

FIG. 6 is a schematic diagram of another eddy current mechanism according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

The technical solution of the present disclosure will be clearly and completely described below in conjunction with the drawings. Obviously, the described embodiments are part of the embodiments of the present disclosure, but not all of them. Based on the embodiments of the present disclosure, all other embodiments obtained by those skilled in the art without exerting creative effort fall within the scope of protection of the present disclosure.

In the description of the present disclosure, it should be noted that the terms "above," "below" for direction description are defined according to the direction in which the test device under normal use conditions. The terms "first," "second," "third," "fourth," and "fifth" are merely used for the purpose of illustration and are not to be understood to indicate or imply relative importance. Further the technical features involved in the different embodiments of the present disclosure described below may be combined with each other as long as they do not conflict with each other.

FIG. 1 is a structural schematic diagram of a small-sized fast cold and hot shock test device according to some embodiments of the present disclosure. FIG. 2 is a structural schematic diagram of a section structure of a test head according to some embodiments of the present disclosure. FIG. 3 is a structural schematic diagram of an adjusting device according to some embodiments of the present disclosure. As shown in FIG. 1, the small-sized fast cold and hot shock test device (referred to as test device) may include a host 1, a test head 2, and an adjustment device 3.

The host 1 may be used to collect, analyze, and/or process data and issue a control command to realize a control function. In some embodiments, the host 1 includes a control device at least used to send temperature control data to the test head 2.

The test head 2 refers to a structure used to provide test space and perform test. In some embodiment, the test head may be used for cold and hot shock to a component under test by temperature control and output of compressed air. The compressed air may be air through compression, and the compressed air may be provided by an air supply system.

The adjustment device **3** is used to adjust a position of the test head **2**. More descriptions regarding the adjustment device **3** may be found below.

In some embodiment, as shown in FIG. **2**, the test head **2** includes an eddy current mechanism **21** for cooling or heating the compressed air, a heater **22**, a nozzle **23**, etc.

The eddy current mechanism **21** is used to change a temperature of the compressed air, such as to cool or heat the compressed air. In some embodiments, the eddy current mechanism **21** may be used to divide the compressed air. For example, the compressed air is divided to form cold air and hot air. The eddy current mechanism **21** has an air inlet **211**, a cold air outlet **212**, and a hot air outlet **213**. The air inlet **211** of the eddy current mechanism **21** is connected with an air supply system through an intake air line unit **27**. The cold air outlet **212** is connected with the air inlet end of the heater **22**.

The heater **22** is used to further heat the compressed air. In some embodiment, the heater **22** includes an air inlet end and an air outlet end.

The nozzle **23** refers to a structure for guiding a fluid (e.g., compressed air) to eject in a certain direction and shape. The nozzle **23** is connected with the air outlet end of the heater **22**. The nozzle **23** may form air used for shock by guiding the compressed air in the heater **22** to be sprayed outward.

In some embodiment, the test device is supplied with the compressed air by the air supply system. The compressed air is passed through the host **1** first, and then input into the eddy current mechanism **21** of the test head **2**, which makes the airflow supplied to the eddy current mechanism **21** more stable. In some embodiment, the air supply system refers to a device or system that can provide the compressed air such as a central air compression system at a production and manufacturing site. The compressed air simultaneously generates a cold air flow and a hot air flow after passing through the eddy current mechanism **21**. The temperature of the cool air and the hot air may be preset. The cool air is fed into the heater **2** from the cold air outlet **212**, and is compensated by the heater **2** to adjust the temperature of the cool air to a required temperature, and then sprayed out of the nozzle **23**. That is, the temperature of the compressed air may be mainly regulated by the eddy current mechanism **21** and the heater **22**. In some embodiment, the temperature of the compressed air output by the test device is range from -55°C . to 220°C ., and the test temperature range is wide. In some embodiment, the speed of heating and cooling of the test device is fast, which only takes about 1 min to reduce from the highest temperature of 220°C . to the lowest temperature of -55°C ., and only takes about 20 s to rise from the lowest temperature of -55°C . to 220°C . The time error of the extreme temperature change within the temperature range does not exceed 5 s. Merely by way of an example, for the most commonly used temperature in the test, for example, it only takes about 12 s for the temperature to reduce from 70°C . to 0°C ., and only takes about 20 s for the temperature to reduce from 85°C . to -40°C ., and the time error of temperature rise and reduction does not exceed 2 s~3 s, which greatly improves the efficiency compared with the existing cold and hot shock test device.

In some embodiments, since equipped with the eddy current mechanism **21**, the test device does not need to be equipped with a compressor and a semiconductor refrigeration sheet, so as to reduce an overall volume of the device and make the device easy to move. At the same time, the eddy current mechanism does not need electricity for heating and cooling compressed air, which greatly reduces energy consumption compared with the existing cold and

heat shock test device equipped with the compressor. In some embodiments, the test head **2** is designed to make the test device suitable for situations where selective testing is required. For example, a situation where a component on a circuit board is tested without causing damage to other surrounding components.

The intake line unit **27** refers to a pipeline used to guide the compressed air into the eddy current mechanism **21**.

FIG. **4** is a schematic diagram of an eddy current mechanism according to some embodiments of the present disclosure. In some embodiments, as shown in FIG. **4**, the intake air line unit **27** may be connected with an air supply system (not shown) and the air inlet **211** of the eddy current mechanism **21**.

In some embodiments, as shown in FIG. **4**, a bypass line unit **26** is arranged between the eddy current mechanism **21** and the heater **22**. The bypass line unit **26** includes a first line **261** with one end connected with the hot air outlet **213**, a first solenoid valve **262** disposed on the first line **261**, a second line **263** with one end connected with the first line **261** and the other end connected with the air inlet end of the heater, and a second solenoid valve **264** disposed on the second line **263**, and the connection of the one end of the second line **263** on the first line **261** is between the first solenoid valve **262** and the air inlet end of the heater **22**.

In some embodiments, the small-sized fast cold and hot shock test device has a cooling working state and a heating working state. When the small-sized fast cold and hot shock test device is in the cooling working state, the first solenoid valve **262** is opened and the second solenoid valve **264** is closed. At this time, the hot air generated by the eddy current mechanism **21** is output from the hot air outlet **213**, and the cold air is output from the cold air outlet **212** and into the heater **22** downstream. When the small-sized fast cold and hot shock test device is in the heating working state, the first solenoid valve **262** is closed and the second solenoid valve **264** is opened. At this time, the cold air generated by the eddy current mechanism **21** is output from the cold air outlet **212** and into the heater **22** downstream, and the hot air enters the heater **22** downstream through the second line **263**. A final function of the hot air entering the heater **22** is to compensate the temperature of compressed air and reduce the heating burden of the heater **22**, thus, further improving the heating efficiency and reducing the energy consumption of the heater **22**.

In some embodiments, other connection manners between the eddy current mechanism **21**, the heater **22**, and the air supply system may be provided. More descriptions regarding the other connection manners may be found in FIG. **5**, FIG. **6**, and the relevant descriptions thereof.

In some embodiments, as shown in FIG. **2**, the test head **2** further includes a shell **24**. The shell **24** serves as a mounting basis of a component in the test head **2** and protects the component at the same time. In some embodiments, the eddy current mechanism **21** and the heater **22** are set in the shell **24**. In some embodiments, a temperature sensor (not shown) is set in the nozzle **22**. The temperature sensor is used to sense the temperature of the compressed air entering the nozzle **22** from the heater **22** and transmitting sensing data to the control device.

In some embodiments, the test head **2** further includes a cover **25** set at a bottom of the shell **24**. During the test, the cover **25** covers the component under test to form a closed space isolated from external air to make the air used for shock more concentrated and the ambient temperature in the closed space closer to the required temperature for testing,

which enables more accurate control of the temperature of the test environment and is conducive to energy saving.

In some embodiments, the cover 25 may be made of a thermal insulation material, such as fiberglass or asbestos.

In some embodiments, the cover 25 includes a cavity 251 for accommodating the component during the test, a first cavity 252 above the cavity 251, and a second cavity 253 around an outside of the cavity 251.

In some embodiments, an outer part of the first cavity 252 and an upper part of the second cavity 253 are connected to form a sealing space sealed and isolated from the cavity 251. The sealing space has an inlet 254 for air input and an outlet for air output, and the cavity 251 has an air outlet 255 for air output. The nozzle 23 passes through the first cavity 252 and a bottom of the nozzle 23 extends into the cavity 251. Frost or mist may form on the wall of the cavity 251 when the extremely low temperature (e.g., -55°C .) airflow meets the external normal temperature airflow during the cold and hot shock test, which is not conducive to an operator observing the real-time test situation of the component under test. Therefore, the inlet 254 and the outlet are set in order to continuously input air into the sealing space during the test, which makes the air flow in the second cavity 253 to form an air curtain and effectively prevents the condensation of water mist and frost on the wall of the cavity 251 under the action of air flow, so that it is convenient for the operator to observe a state of the component under test directly, and the test can be carried out normally.

In some embodiments, the first cavity 252 may be formed by connecting a first body 252-1, a second body 252-2, and a third body 252-3. Positions of the first body 252-1 and the second body 252-2 may correspond up and down. That is, the first body is located directly above the second body. An upper end and a lower end of the third body 252-3 are connected with the first body 252-1 and the second body 252-2, respectively. In some embodiments, the second cavity 253 is formed by connecting a gasket 253-1, a fourth body 253-2, and a fifth body 253-3. An upper end and a lower end of the fourth body 253-2 are connected with a lower end of the second body 252-2 and the gasket 253-1, respectively, and an upper end and a lower end of the fifth body 253-3 are connected with the lower end of the second body 252-2 and the gasket 253-1, respectively.

In some embodiments, a plurality of holes may be provided in the second body 252-2 to connect the first cavity 252 with the second cavity 253, so that the first cavity 252 and the second cavity 253 may form the sealing space. In some embodiments, the cavity 251 may be formed by connecting the second body 252-2 and the fifth body 253-3. In some embodiments, the inlet 254 and the air outlet 255 are both set on the third body 252-3.

In some embodiments, the gasket 253-1 is set to make an interior of the cover 25, including the cavity 251 and the sealing space composed of the first cavity 252 and the second cavity 253, more enclosed.

In some embodiments, the shell 24 may be fixedly connected with the first body 252-1 of the cover 25.

In some embodiments, the inlet 254 and the outlet can be set at any height on the third body 252-3, and can also be set on the fourth body 253-2. On the premise that water mist and frost can be prevented from condensation on the wall of cavity 251, the setting positions and setting heights of the inlet 254 and the outlet may be determined as needed. In some embodiments, the outlet may be a connection gap between the gasket 253-1 and the fourth body 253-2. On the premise that the air inside the cavity 251 can be discharged, the setting position and setting mode of the air outlet 255

may be determined as needed. In some embodiments, the air outlet 255 may be set on the third body 252-3, and the second body 252-2 is provided with vents (not shown). The air outlet 255 is connected with the vents through a connecting pipe (not shown) located in the first cavity 252, and the air in the cavity 251 passes through the connecting pipe and is then discharged from the air outlet 255.

In some embodiments, the test device may adjust the position of the test head 2 through the adjusting device 3. In some embodiments, the adjusting device 3 is not attached to the test device and can be installed in any position such as table top, floor, etc. By changing the position of the adjusting device 3, a test area covered by the test head 2 movement is enlarged to a certain extent, and the operation convenience and mobility flexibility of the test device are effectively improved.

In some embodiments, as shown in FIG. 3, the adjustment device 3 may include a fixing bracket 31 and an adjusting bracket 32 capable of being expanded and folded, and one end of the adjusting bracket is connected with the fixing bracket 31, and the other end of the adjusting bracket is slidably connected with the test head 2. Being expanded and folded of the adjusting bracket 32 enables the test head 2 to move over a wider area.

In some embodiments, the adjusting bracket includes a sliding block 321 slidably connected with the test head 2, a connecting block 322 rotationally connected with the sliding block 321, a first connecting arm 323 rotationally connected with the connecting block 322, and a second connecting arm 324 rotationally connected with the first connecting arm 323 and the fixing bracket 31.

In some embodiments, the sliding connection between the test head 2 and the sliding block 321 enables the test head 2 and the sliding block 321 to slide along a length direction of the test head 2, so that a relative height of the test head 2 and the adjusting bracket 32 can be adjusted. An angle of the sliding block 321 relative to the connecting block 322 is adjusted through the rotary connection between the sliding block 321 and the connecting block 322, so as to change the angle of the test head 2 relative to the adjusting bracket 32. The rotary connection of the first connecting arm 323 and the second connecting arm 324 realizes the expanding and folding of the adjusting bracket 32, and increases the test area that covered by the test head 2 movement. The second connecting arm 324 is rotationally connected with the fixed bracket 31, so that the adjusting bracket 32 can arbitrarily change its direction around the fixing bracket 31 in the case that the fixing bracket 31 has been connected to other objects.

In some embodiments, a bottom of the fixing bracket 31 is provided with a clamping mechanism 311 for fixing a position of the adjusting device 3. The clamping mechanism 311 is separably connected to other objects. The clamping mechanism 311 can be clamped on the table or other objects, and can also be clamped on the strip or plate shaped components of other devices, which is not only suitable for a plurality of objects, but also can be repeatedly disassembled. In some embodiments, as shown in FIG. 3, the clamping mechanism 311 includes a holder 311-1 connected with the bottom of the fixing bracket 31 and an adjusting knob 311-2 arranged on the holder 311-1. The holder 311-1 is used to connect with other object such as a table, and the adjusting knob 311-2 can adjust a clamping limit within a certain range, so that the clamping mechanism 311 can be applied to the objects with different thickness. In practical application, the clamping mechanism 311 can also adopt any

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other feasible structure that may achieve the clamping function according to the actual situation.

FIG. 5 is a schematic diagram of another eddy current mechanism according to some embodiments of the present disclosure. In some embodiments, as shown in FIG. 5, the intake air line unit 27 may be connected with the air supply system and the air inlet end of the heater 22. The intake air line unit 27 includes a third line 271 with one end connected with the air inlet 211 and the other end connected with the air supply system, a third solenoid valve 272 disposed on the third line 271, a fourth line 273 with one end connected with the air inlet end of the heater 22 and the other end connected with the air supply system, and a fourth solenoid valve 274 disposed on the fourth line 273.

In some embodiments, the small-sized fast cold and hot shock test device has a cooling working state and a heating working state. When the small-sized fast cold and hot shock test device is in the cooling working state, the third solenoid valve 272 is opened and the fourth solenoid valve 274 is closed. Compressed air enters the eddy current mechanism 21 through the third line 371, the hot air generated by the eddy current mechanism 21 is output from the hot air outlet 213, and the cold air is output from the cold air outlet 212 and then into the heater 22 downstream. When the small-sized fast cold and hot shock test device is in the heating working state, the third solenoid valve 272 is closed and the fourth solenoid valve 274 is opened. The compressed air enters the heater 22 directly for heating treatment through the fourth line 273. At this time, the compressed air entering the heater 22 is normal temperature, which takes less time and consumes less energy than the heater 22 heats up the compressed air to the target temperature than that of heating up the cold air to the target temperature.

FIG. 6 is a schematic diagram of another eddy current mechanism according to some embodiments of the present disclosure. In some embodiments, as shown in FIG. 6, the bypass line unit 26 is arranged between the eddy current mechanism 21 and the heater 22, and the intake air line unit 27 is connected with the air supply system and the air inlet end of the heater 22. More descriptions regarding the bypass line unit 26 may be found in FIG. 4 and the relevant descriptions thereof. More descriptions regarding the intake air line unit 27 may be found in FIG. 4 and the relevant descriptions thereof.

In some embodiments, the small-sized fast cold and hot shock test device has a cooling working state, a heating working state, and a high temperature heating state. When the small-sized fast cold and hot shock test device is in the cooling working state, the first solenoid valve 262 and the third solenoid valve 272 are opened, and the second solenoid valve 264 and the fourth solenoid valve 274 are closed. Compressed air enters the eddy current mechanism 21 through the third line 371, the hot air generated by the eddy current mechanism 21 is output from the hot air outlet 213, the cold air is output from the cold air outlet 212 and then into the heater 22 downstream. When the small-sized fast cold and hot shock test device is in the heating working state, the second solenoid valve 264 and the third solenoid valve 272 are opened, and the first solenoid valve 262 and the fourth solenoid valve 274 are closed. The compressed air enters the eddy current mechanism 21 through the third line 371, and the hot air generated by the eddy current mechanism 21 enters the heater 22 downstream through the second line 263 to compensate temperature. When the small-sized fast cold and hot shock test device is in the high temperature heating state, the fourth solenoid valve 274 is opened, and the first solenoid valve 262, the second solenoid valve 264, and the

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third solenoid valve 272 are all closed. The compressed air enters the heater 22 directly for heating treatment through the fourth line 273.

It should be noted that the heating working state and the high temperature heating state can be selected freely according to the test requirements. When the required air for test is hot air at a lower target temperature, the heating working state can be selected, and the cold air and the hot air generated by the eddy current mechanism 21 both enter the heater 22, due to the lower target temperature, the burden of the heater 22 by heating the cold air is light, at the same time, the hot air compensates the temperature, which further improves the heating speed and reduces the energy consumption of the heater 22.

In some embodiments, when the required air for test is hot air at a relatively high target temperature, the high temperature heating state can be selected, and the compressed air enters the heater 22 to be heated. At this time, no compressed air enters the eddy current mechanism 21, so that no cold air is generated, and the compressed air is heated to the target temperature from normal temperature, which reduces the heating burden of the heater and saves energy.

The above embodiments are only for illustrating the technical concept and characteristics of the present disclosure, which aims to enable those familiar with the technique to understand the contents of the present disclosure and implement it accordingly, and cannot limit the scope of protection of the present disclosure. Any equivalent changes or modifications made according to the spirit of the present disclosure should be covered within the scope of protection of the present disclosure.

What is claimed is:

1. A cold and hot shock test device, comprising a host, a test head used for cold and hot shock to a component under test by temperature control and output of compressed air, and an adjustment device for adjusting a position of the test head, wherein

the host includes a control device at least used to send temperature control data to the test head, and the test head includes an eddy current mechanism for cooling or heating the compressed air, a heater including an air inlet end and an air outlet end, and a nozzle, wherein the eddy current mechanism includes an air inlet, a cold air outlet, and a hot air outlet, the air inlet is connected with an air supply system through an intake air line unit, the cold air outlet is connected with the air inlet end of the heater, and the nozzle is connected with the air outlet end of the heater.

2. The cold and hot shock test device according to claim 1, wherein a bypass line unit is arranged between the eddy current mechanism and the heater, the bypass line unit includes a first line with one end connected with the hot air outlet, a first solenoid valve disposed on the first line, a second line with one end connected with the first line and the other end connected with the air inlet end of the heater, and a second solenoid valve disposed on the second line, and the connection of the one end of the second line on the first line is between the first solenoid valve and the air inlet end of the heater.

3. The cold and hot shock test device according to claim 2, wherein

the cold and hot shock test device has a cooling working state and a heating working state, the first solenoid valve is opened and the second solenoid valve is closed when the cold and hot shock test device is in the cooling working state; and the first solenoid valve is

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closed and the second solenoid valve is opened when cold and hot shock test device is in the heating working state.

4. The cold and hot shock test device according to claim 2, wherein the intake air line unit is further connected with the air inlet end of the heater, the intake air line unit includes a third line with one end connected with the air inlet and the other end connected with the air supply system, a third solenoid valve disposed on the third line, a fourth line with one end connected with the air inlet end of the heater and the other end connected with the air supply system, and a fourth solenoid valve disposed on the fourth line.

5. The cold and hot shock test device according to claim 4, wherein

the third solenoid valve is opened and the fourth solenoid valve is closed when the cold and hot shock test device is in the cooling working state and the heating working state.

6. The cold and hot shock test device according to claim 4, wherein

the cold and hot shock test device further has a high temperature heating state, and the fourth solenoid valve is opened, and the first solenoid valve, the second solenoid valve, and the third solenoid valve are all closed when the cold and hot shock test device is in the high temperature heating state.

7. The cold and hot shock test device according to claim 1, wherein the test head further includes a shell that the eddy current mechanism and the heater are set in, the nozzle is provided with a temperature sensor for sensing the temperature of the compressed air entering the nozzle from the heater and transmitting sensing data to the control device.

8. The cold and hot shock test device according to claim 7, wherein the test head further includes a cover set at a bottom of the shell, the cover being used to cover the component under test to form a closed space isolated from external air during the test.

9. The cold and hot shock test device according to claim 8, wherein the cover has a cavity for accommodating the component under test during the test, a first cavity above the cavity, and a second cavity around an outside of the cavity, wherein

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an outer part of the first cavity and an upper part of the second cavity are connected to form a sealing space sealed and isolated from the cavity, the sealing space has an inlet for air input and an outlet for air output, and the cavity has an air outlet for air output; and the nozzle passes through the first cavity and a bottom of the nozzle extends into the cavity.

10. The cold and hot shock test device according to claim 9, wherein the first cavity is formed by connecting a first body, a second body, and a third body, the first body is located directly above the second body, and an upper end and a lower end of the third body are connected with the first body and the second body, respectively;

the second cavity is formed by connecting a gasket, a fourth body, and a fifth body, an upper end and a lower end of the fourth body are connected with a lower end of the second body and the gasket, respectively, and an upper end and a lower end of the fifth body are connected with the lower end of the second body and the gasket, respectively; and

the cavity is formed by connecting the second body and the fifth body; and the inlet and the air outlet are set on the third body.

11. The cold and hot shock test device according to claim 1, wherein the adjustment device includes a fixing bracket and an adjusting bracket capable of being expanded and folded, one end of the adjusting bracket being connected with the fixing bracket, the other end of the adjusting bracket being slidably connected with the test head, and a bottom of the fixing bracket being provided with a clamping mechanism for fixing a position of the adjusting device.

12. The cold and hot shock test device according to claim 11, wherein the adjusting bracket includes a sliding block connected with the test head, a connecting block rotationally connected with the sliding block, a first connecting arm rotationally connected with the connecting block, and a second connecting arm rotationally connected with the first connecting arm and the fixing bracket.

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